Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 6 Handout

Tuesday, September 28

Problem 6.1

You have been hired to investigate a disturbing connection between ice cream sales and crime. You are given a report that includes the joint distribution of random variable $S \in \{\texttt{low}, \texttt{high}\}$ that represents ice cream sales and random variable $X \in \{\texttt{low}, \texttt{high}\}$ that represents the crime rate in the city.

s	x	$p_{S,X}(s,x)$
low	low	0.15
low	high	0.15
high	low	0.3
high	high	0.4

(a) Are random variables S and X independent?

(b) Assume that the trends of ice cream sales and crime during any day are independent of all other days. Suppose the ice cream sales were low on Monday and high on Tuesday. On which day is the crime rate more likely to be high?

(c) After further investigation, you obtain additional information about the dependencies of the ice cream sales and crime volume on the temperature on that day. Let $T \in \{\texttt{low}, \texttt{high}\}$ denote the temperature, and assume that the conditional distributions of S and X given T are as follows:

s	x	$p_{S,X\|T}(s,x\|\texttt{low})$
low	low	0.36
low	high	0.24
high	low	0.24
high	high	0.16

s	x	$p_{S,X\|T}(s,x\|\texttt{high})$
low	low	0.08
low	high	0.12
high	low	0.32
high	high	0.48

Are random variables S and X conditionally independent given T?

(d) Sort the quantities below in the increasing order:

$$I(S;X|T) \qquad I(X;S) \qquad I(S;X|T=\texttt{high}).$$

(e) Can you determine the distribution $p_T(\cdot)$ from the tables above? If so, find it. If not, explain what additional information is needed to determine $p_T(\cdot)$.